OOP Assignment1: Designing Interacting Objects

Objective

The purpose of this assignment is to practice designing software from an object-oriented perspective. You will organize a problem into classes, identify relationships between those classes, decide which behaviors should be shared or overridden, and develop interactions between objects.

For this assignment, choose a scenario that interests you and model part of it using object-oriented programming. Your idea can come from a game, movie, anime, book, real-world situation, or an application you would like to design. You do not need to model an entire system. Focus on one part where different types of objects interact and where there are meaningful similarities and differences between them. Your objective is to translate that scenario into an object-oriented design, implement the interaction, and explain the reasoning behind your design choices.

Part 1 - Design (20 points)

Identify the main classes in your system and determine how they are related. Your design should include a meaningful inheritance relationship. Decide what information and behavior belong in a parent class and what should be specialized in subclasses. For example, a general `Character` class contains properties such as `name`, `health`, and `attack_power`, along with behaviors that are common to all characters as we discussed during the lecture. Specific character classes could inherit those features while overriding a method like `spcial_attack()` to provide different behavior. Describe these relationships and justify them in your report.

Part 2 - Implement the Interaction (40 points)

The objects in your program should interact with one another rather than simply store and display information. Remember objects by themselves do not coordinate or trigger actions. It provides the capabilities of the character/or the object. You probably need an additional class that coordinates the interaction between the objects like the Game class presented in the class. This coordinator class is the one should trigger the behavior of individual objects and communicate between them. In this case, the coordinator class will capture the entry point of your program.

Part 3 - Project Presentation (20 points)

Prepare a presentation that introduces your scenario and explains the object-oriented design behind your solution. Begin by describing the scene, application, or situation you selected so that the class understands what you are modeling. Then explain how you translated that scenario into classes and relationships.

Your presentation should explain the important design decisions you have made. Discuss what belongs in the parent class, what is specialized or overridden in the subclasses, and how the objects interact. Walk through at least one important interaction in your program and explain how the classes work together to produce that behavior. The purpose of the presentation is to communicate your design decisions and reasoning and to onboard your classmates so they can help edit or support your application.

Part 4 - Draft a Final Project Proposal (20 points)

Use this assignment as a starting point for thinking about your final project. Prepare a draft proposal for an application, game, simulation, or other software project that you would be interested in developing. Your proposal should describe the main idea of the project, what the program is intended to do, and the major components you expect it to contain. Identify some initial classes or objects that may be part of the project and describe how they might interact. You should also identify possible opportunities to use concepts such as inheritance, interfaces, composition, or polymorphism. This is an initial proposal. Your design is expected to evolve as you continue learning new concepts and developing the project.

File Submission

Submit your assignment materials in a clearly organized folder or compressed file. Your submission should include:

- **Report:** description of your system and its elements and their purpose and the coordination between them, mention any challenges you faced during the development.
- **Source Code:** All code files required to run your scenario/application.
- **Presentation Slides:** The slides are used to explain your scenario, class design, and object interactions.
- **Final Project Proposal:** Your draft proposal for the final project.
- **References:** Include references for the scene, game, movie, application, images, documentation, tutorials, or other external resources you used.

Use clear file names and organize your files so that each part of the submission is easy to identify.

GitHub Recommendation

You are strongly encouraged to use GitHub to store and manage your work. For this assignment, you may create a repository containing your source code, documentation, and related files. If you use GitHub, include the repository link with your submission.

You should plan to use GitHub for your final project. Use the repository throughout the development process rather than uploading the completed project only at the end. Your commit history should show how the project develops over time as you add features, revise your class design, fix problems, and improve the program. Using GitHub in this way provides a record of your development process and helps demonstrate the progress of your final project.